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# Homework1 - MAS

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## Abstract

This report details the solution and analysis for Homework Assignment 1, focusing on the implementation, comparison, and discussing the performance of three algorithms: SARSA, Q-learning, and Double Q-learning in navigating a maze environment, for the single agent case.

## 1 Introduction

Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards. RL agents are not given explicit instructions on what to do; instead, they discover optimal behaviors by interacting with an environment and receiving feedback in the form of rewards or penalties.

We propose to investigate the performance of three foundational RL algorithms: SARSA, Q-Learning and Double Q-learning. They are applied on two grid-based navigation environments. GridWorld A features static obstacles while Gridworld B introduces stochastic-like dynamics through a crosswind that displaces the agent upward.

Task1 presents a comparison of the three selected temporal-difference learning algorithms across both gridworld configurations. This evaluation analyzes the impact of hyperparameter tuning—specifically the exploration rate ( $\epsilon$ ) and the learning rate ( $\alpha$ )—on key performance metrics, including convergence time, optimal path efficiency, and algorithmic robustness.

Task 2 increases the complexity in the windy environment (Gridworld B) by expanding the agent's action space to an eight-way movement scheme that includes diagonals. We only test this expanded setup using the SARSA algorithm to see if the extra movement options improve performance.

## 2 Algorithms

**SARSA** is an on-policy algorithm: it learns the value of the policy it is following, including exploration. At each step it observes the next state and selects the next action using the same -greedy policy, then updates using that actual next action. It accounts for the cost of exploration in its Q-values.

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)]$$

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where  $a'$  is chosen by the -greedy policy from  $s'$ .

**Q-learning** is off-policy: it always updates toward the best possible next action, regardless of what action the agent actually takes. The target uses  $\max_{a'} Q(s', a')$ , so Q-Learning learns the optimal policy even while behaving greedily with some exploration. This makes it more aggressive than SARSA but can overestimate values in stochastic environments.

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[ r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

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31 **Double Q-Learning** Q-Learning has maximization bias: using the same Q-table to both select  
 32 and evaluate the best action leads to overestimation of values. Double Q-Learning fixes this by  
 33 maintaining two independent tables  $Q_1$  and  $Q_2$ . At each step, one table selects the best action and  
 34 the other evaluates it, with the role of each table chosen randomly. This reduces overestimation.

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha \left[ r + \gamma Q_2(s', \arg \max_{a'} Q_1(s', a')) - Q_1(s, a) \right]$$

35 (symmetrically for  $Q_2$ , with roles swapped)

$$Q_2(s, a) \leftarrow Q_2(s, a) + \alpha \left[ r + \gamma Q_1(s', \arg \max_{a'} Q_2(s', a')) - Q_2(s, a) \right]$$

### 36 3 GridWorld A - Task1

37 To study the effect of hyperparameters, all three algorithms were run across every combination of  
 38 three learning rates ( 0.1, 0.3, 0.5) and three exploration rates ( 0.01, 0.1, 0.3), yielding 27 training  
 39 runs in total. Performance was assessed using three metrics: convergence time (the first episode at  
 40 which the 50-episode rolling average of steps dropped within 30 of the minimum), path efficiency  
 41 (average steps over 100 greedy evaluation trials after training), and robustness (standard deviation of  
 42 path efficiency across 10 independent random seeds).

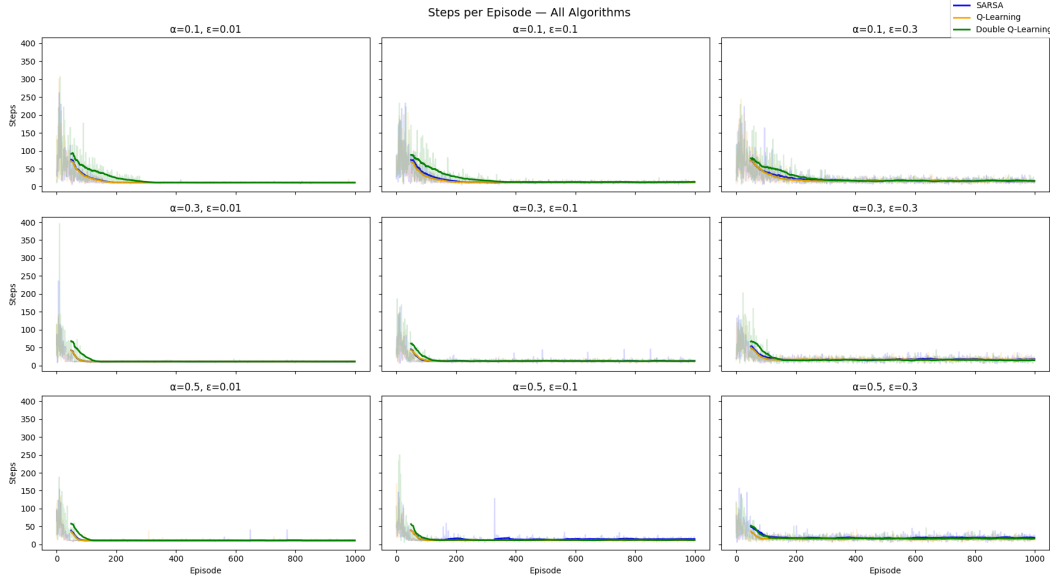


Figure 1: Learning curves (steps per episode) for Gridworld A across all combinations of  $\alpha$  and  $\epsilon$ .

43 All three algorithms start with large negative rewards (around 200 to 400) due to frequent obstacle  
 44 hits and long wandering paths in the early episodes, then steadily climb toward 0 as they learn to  
 45 reach the goal without hitting obstacles. The rate of improvement is clearly driven by : at =0.1 (top  
 46 row) the curves rise slowly and are still improving near episode 1000, while at =0.5 (bottom row) all  
 47 algorithms converge within roughly the first 200 episodes. Higher (right columns) introduces more  
 48 variance in the raw episode rewards but the three algorithms remain closely grouped throughout, with  
 49 no single algorithm consistently dominating in terms of final reward level.

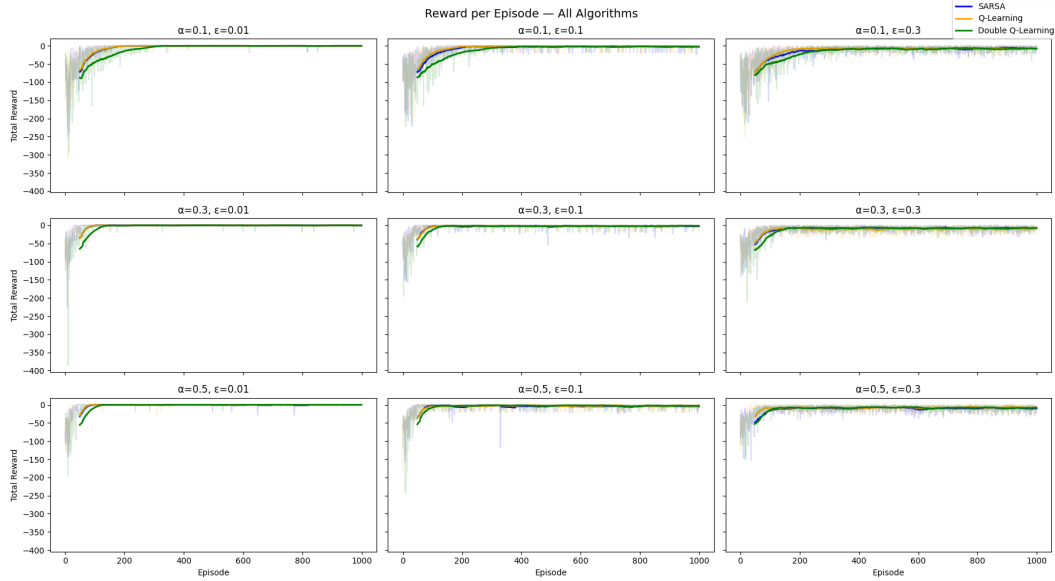


Figure 2: Learning curves (reward per episode) for Gridworld A across all combinations of  $\alpha$  and  $\epsilon$ .

Steps begin high (up to 300–400 in the first episodes) and fall sharply as agents learn a path to the goal. Again  $\alpha$  is the dominant factor: higher learning rates produce a steeper initial drop and a lower plateau. At  $\alpha=0.5$  with low  $\epsilon$  the smoothed curves for all three algorithms collapse to near-optimal within 100–200 episodes, while at  $\alpha=0.1$  convergence is gradual and extends well beyond episode 500. At higher  $\epsilon$  ( $=0.3$ , right column) the smoothed step curves plateau slightly higher and show more residual noise, since the agent keeps exploring even after learning a good policy, occasionally triggering the obstacle penalty and taking longer paths.

Table 1: Convergence Time (episode where 50-episode rolling average first drops to near-optimal) – Gridworld A

| Algorithm         | $\alpha$ | $\epsilon = 0.01$ | $\epsilon = 0.1$ | $\epsilon = 0.3$ |
|-------------------|----------|-------------------|------------------|------------------|
| SARSA             | 0.1      | 91                | 100              | 117              |
|                   | 0.3      | 58                | 60               | 74               |
|                   | 0.5      | 51                | 56               | 77               |
| Q-Learning        | 0.1      | 85                | 90               | 105              |
|                   | 0.3      | 56                | 60               | 69               |
|                   | 0.5      | 50                | 53               | 52               |
| Double Q-Learning | 0.1      | 159               | 164              | 174              |
|                   | 0.3      | 79                | 76               | 98               |
|                   | 0.5      | 69                | 65               | 72               |

Table 2: Path Efficiency (average steps over 100 greedy trials) – Gridworld A

| Algorithm         | $\alpha$ | $\epsilon = 0.01$ | $\epsilon = 0.1$ | $\epsilon = 0.3$ |
|-------------------|----------|-------------------|------------------|------------------|
| SARSA             | 0.1      | 11.00             | 11.00            | 11.00            |
|                   | 0.3      | 11.00             | 11.00            | 13.00            |
|                   | 0.5      | 11.00             | 13.00            | 500.00           |
| Q-Learning        | 0.1      | 11.00             | 11.00            | 11.00            |
|                   | 0.3      | 11.00             | 11.00            | 11.00            |
|                   | 0.5      | 11.00             | 11.00            | 11.00            |
| Double Q-Learning | 0.1      | 11.00             | 11.00            | 11.00            |
|                   | 0.3      | 11.00             | 11.00            | 11.00            |
|                   | 0.5      | 11.00             | 11.00            | 11.00            |

Table 3: Robustness (average steps and standard deviation across 10 random seeds) – Gridworld A

| Algorithm         | $\alpha$ | $\varepsilon = 0.01$ |      | $\varepsilon = 0.1$ |        | $\varepsilon = 0.3$ |        |
|-------------------|----------|----------------------|------|---------------------|--------|---------------------|--------|
|                   |          | Avg                  | Std  | Avg                 | Std    | Avg                 | Std    |
| SARSA             | 0.1      | 11.00                | 0.00 | 11.00               | 0.00   | 11.00               | 0.00   |
|                   | 0.3      | 11.00                | 0.00 | 12.20               | 0.98   | 12.40               | 0.92   |
|                   | 0.5      | 11.40                | 0.80 | 61.50               | 146.17 | 110.60              | 194.70 |
| Q-Learning        | 0.1      | 11.00                | 0.00 | 11.00               | 0.00   | 11.00               | 0.00   |
|                   | 0.3      | 11.00                | 0.00 | 11.00               | 0.00   | 11.00               | 0.00   |
|                   | 0.5      | 11.00                | 0.00 | 11.00               | 0.00   | 11.00               | 0.00   |
| Double Q-Learning | 0.1      | 11.00                | 0.00 | 11.00               | 0.00   | 11.00               | 0.00   |
|                   | 0.3      | 11.00                | 0.00 | 11.20               | 0.60   | 11.20               | 0.60   |
|                   | 0.5      | 11.60                | 0.92 | 11.20               | 0.60   | 11.20               | 0.60   |

Q-Learning converges fastest across all hyperparameter settings (50–105 episodes), followed by SARSA (51–117 episodes), while Double Q-Learning is the slowest to converge (65–174 episodes) a consequence of splitting updates across two tables, requiring more samples to stabilize. In terms of path efficiency, Q-Learning and Double Q-Learning consistently reach the optimal 11-step path regardless of  $\alpha$  and  $\varepsilon$ , while SARSA degrades at high learning rate and high exploration (reaching 500 steps at  $\alpha=0.5, \varepsilon=0.3$ ), because its on-policy nature causes it to overfit to the exploratory behavior. Robustness reveals: Q-Learning achieves zero standard deviation across all 10 seeds and all hyperparameter combinations, making it the most reliable algorithm in this environment; Double Q-Learning is nearly as stable with only small deviations at high  $\alpha$ ; SARSA becomes highly unstable at  $\alpha=0.5$  combined with 0.1, with standard deviations exceeding 146, indicating that aggressive learning rates amplify the sensitivity of the on-policy update to random initialization.

#### 4 GridWorld B - Task1

All three algorithms were evaluated across every combination of three learning rates ( $\alpha = 0.1, 0.3, 0.5$ ) and three exploration rates ( $\varepsilon = 0.01, 0.1, 0.3$ ), yielding 27 training runs per algorithm. The same three metrics were used: convergence time, path efficiency over 100 greedy evaluation trials, and robustness measured as standard deviation of path efficiency across 10 independent random seeds.

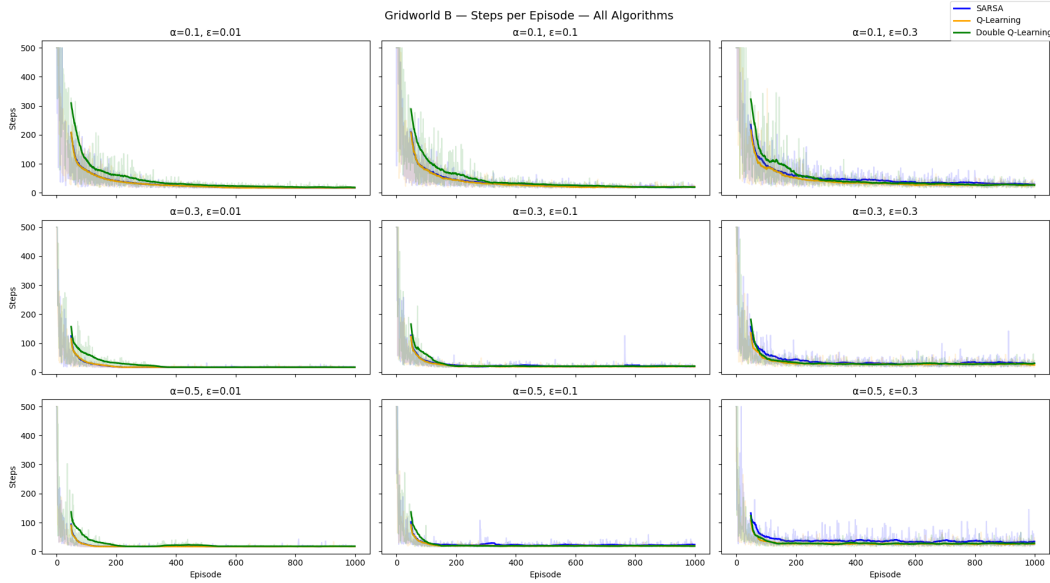


Figure 3: Learning curves (steps per episode) for Gridworld B across all combinations of  $\alpha$  and  $\varepsilon$ .

All algorithms begin with strongly negative rewards (down to 500) as agent wander extensively before finding the goal, then climb toward 0 as it learns to navigate the wind. Compared to Gridworld

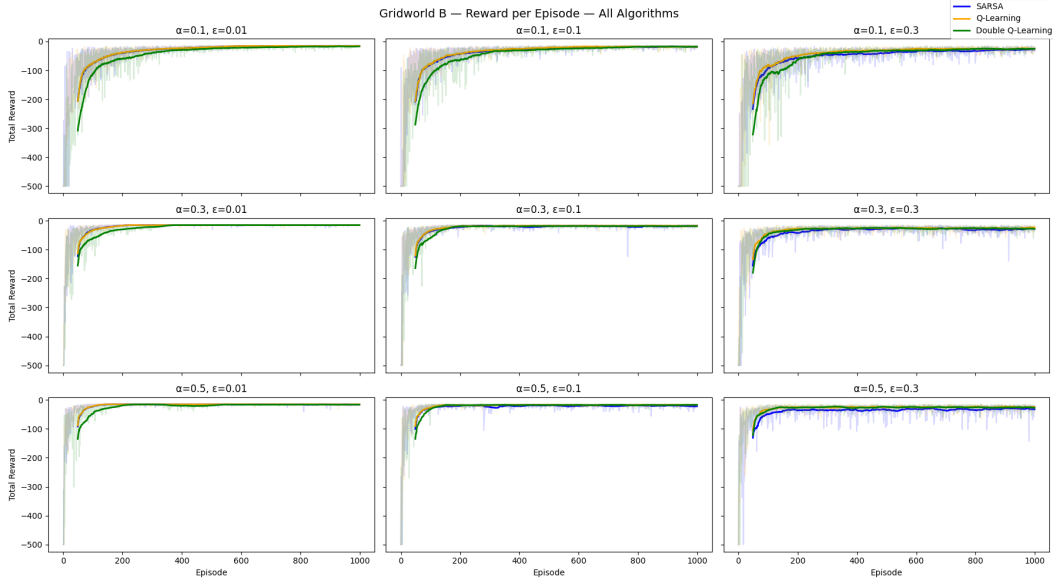


Figure 4: Learning curves (reward per episode) for Gridworld B across all combinations of  $\alpha$  and  $\epsilon$ .

75 A, convergence is noticeably slower — even at  $\alpha=0.5$  the curves are still rising past episode 200 —  
 76 because the wind makes the environment harder to predict and the reward signal is weaker. At high  
 77  $\epsilon=0.3$  (right column), SARSA (blue) plateaus slightly lower than Q-Learning and Double Q-Learning  
 78 because its on-policy updates absorb the cost of continued exploration, while the off-policy methods  
 79 converge to a cleaner policy value regardless of exploration noise.

80 Steps start near 500 the cap and fall sharply within the first 50–100 episodes across all settings,  
 81 reflecting how quickly agent discover any path to the goal. Higher  $\alpha$  again accelerates the descent —  
 82 at  $\alpha=0.5$  the drop is nearly immediate — but unlike Gridworld A, the curves do not fully flatten; a  
 83 persistent noise floor remains, especially at high  $\epsilon$ , visible as frequent spikes throughout training. This  
 84 residual variance is caused by the wind stochastically displacing the agent even along a known good  
 85 path, meaning occasional long detours are unavoidable regardless of how well the policy is learned.  
 86 Double Q-Learning (green) is visibly more volatile early on due to slower value stabilization across  
 87 two tables, but converges to the same level as the others.

Table 4: Convergence Time (episode where 50-episode rolling average first drops to near-optimal) – Gridworld B

| Algorithm         | $\alpha$ | $\epsilon = 0.01$ | $\epsilon = 0.1$ | $\epsilon = 0.3$ |
|-------------------|----------|-------------------|------------------|------------------|
| SARSA             | 0.1      | 173               | 186              | 292              |
|                   | 0.3      | 77                | 82               | 144              |
|                   | 0.5      | 64                | 67               | 113              |
| Q-Learning        | 0.1      | 170               | 181              | 222              |
|                   | 0.3      | 83                | 82               | 98               |
|                   | 0.5      | 63                | 67               | 75               |
| Double Q-Learning | 0.1      | 271               | 272              | 277              |
|                   | 0.3      | 135               | 121              | 103              |
|                   | 0.5      | 104               | 85               | 83               |

88 Gridworld B converges slower than Gridworld A across all algorithms due to the weaker reward signal  
 89 and wind stochasticity. Q-Learning and SARSA converge at similar speeds while Double Q-Learning  
 90 lags at low  $\alpha$  but catches up at higher  $\alpha$  — the same trend as Gridworld A, just more pronounced. Path  
 91 efficiency mirrors Gridworld A exactly: Q-Learning always reaches the optimal path (17 steps),  
 92 SARSA breaks at  $\alpha=0.5$ ,  $\epsilon=0.3$ , and Double Q-Learning adds at most 1 extra step in some configurations.  
 93 The most notable difference from Gridworld A is in robustness: Q-Learning again achieves  $\text{std}=0$   
 94 everywhere, but SARSA's instability is significantly worse — the worst-case standard deviation

Table 5: Path Efficiency (average steps over 100 greedy trials) – Gridworld B

| Algorithm         | $\alpha$ | $\varepsilon = 0.01$ | $\varepsilon = 0.1$ | $\varepsilon = 0.3$ |
|-------------------|----------|----------------------|---------------------|---------------------|
| SARSA             | 0.1      | 17.00                | 17.00               | 17.00               |
|                   | 0.3      | 17.00                | 17.00               | 18.00               |
|                   | 0.5      | 17.00                | 19.00               | 500.00              |
| Q-Learning        | 0.1      | 17.00                | 17.00               | 17.00               |
|                   | 0.3      | 17.00                | 17.00               | 17.00               |
|                   | 0.5      | 17.00                | 17.00               | 17.00               |
| Double Q-Learning | 0.1      | 18.00                | 17.00               | 17.00               |
|                   | 0.3      | 17.00                | 18.00               | 18.00               |
|                   | 0.5      | 18.00                | 17.00               | 17.00               |

Table 6: Robustness (average steps and standard deviation across 10 random seeds) – Gridworld B

| Algorithm         | $\alpha$ | $\varepsilon = 0.01$ |      | $\varepsilon = 0.1$ |        | $\varepsilon = 0.3$ |        |
|-------------------|----------|----------------------|------|---------------------|--------|---------------------|--------|
|                   |          | Avg                  | Std  | Avg                 | Std    | Avg                 | Std    |
| SARSA             | 0.1      | 17.00                | 0.00 | 17.40               | 0.80   | 66.00               | 144.67 |
|                   | 0.3      | 17.00                | 0.00 | 17.30               | 0.46   | 17.50               | 0.92   |
|                   | 0.5      | 17.20                | 0.40 | 65.50               | 144.83 | 259.00              | 241.00 |
| Q-Learning        | 0.1      | 17.00                | 0.00 | 17.00               | 0.00   | 17.00               | 0.00   |
|                   | 0.3      | 17.00                | 0.00 | 17.00               | 0.00   | 17.00               | 0.00   |
|                   | 0.5      | 17.00                | 0.00 | 17.00               | 0.00   | 17.00               | 0.00   |
| Double Q-Learning | 0.1      | 17.30                | 0.46 | 17.30               | 0.46   | 17.20               | 0.40   |
|                   | 0.3      | 17.20                | 0.40 | 17.40               | 0.49   | 17.80               | 0.40   |
|                   | 0.5      | 17.50                | 0.67 | 17.50               | 0.67   | 17.40               | 0.49   |

grows from 194 in Gridworld A to 241 in Gridworld B, and new failure modes appear at lower hyperparameter values, confirming that wind amplifies SARSA’s sensitivity to aggressive learning rates and high exploration.

## 5 Task2

Only SARSA was implemented, trained for 1000 episodes across all combinations of three learning rates ( 0.1, 0.3, 0.5) and three exploration rates ( 0.01, 0.1, 0.3), yielding 9 training runs. The same three metrics were computed — convergence time, path efficiency over 100 greedy trials, and robustness across 10 random seeds — and results were compared directly against the 4-action SARSA from Task 1.

Rewards start deeply negative (near 500) and rise toward 0 as SARSA learns to exploit diagonal moves. The convergence shape is similar to 4-action Gridworld B, but the curves climb faster and settle closer to 0 — reflecting the shorter paths the agent discovers with diagonals. At low (left column) the smoothed curve is clean and stable, while at  $\varepsilon=0.3$  (right column) a persistent gap remains below 0 even after 1000 episodes, as the on-policy update absorbs the cost of random diagonal moves that often undo the wind compensation.

Steps fall sharply from 500 within the first 50–100 episodes and plateau at a much lower floor than the 4-action version — visibly settling around 7–20 steps in the best configurations versus 17+ in Task 1. Higher  $\alpha$  accelerates the initial drop, and at  $\varepsilon=0.5$  the curve collapses almost immediately. The key visual difference from 4-action SARSA is the lower steady-state plateau, confirming that diagonal moves allow the agent to find genuinely shorter paths. At  $\varepsilon=0.3$  (right column) the smoothed curve plateaus higher and retains more noise throughout training, since frequent exploration of the larger 8-action space keeps introducing suboptimal diagonal moves that derail a learned policy.

8-action SARSA achieves 7 steps in almost all stable configurations, compared to 17 steps for the 4-action version — a 2.4 $\times$  improvement that represents the theoretical minimum, since the goal is exactly 7 columns away and diagonal moves allow the agent to advance one column per step while letting the wind handle the vertical displacement. The clearest tradeoff appears at high exploration: at  $\varepsilon=0.3$ , 8-action SARSA fails (500 steps) across most values because random exploration over 8 actions is more disruptive than over 4 — a bad random diagonal can undo wind compensation in a single

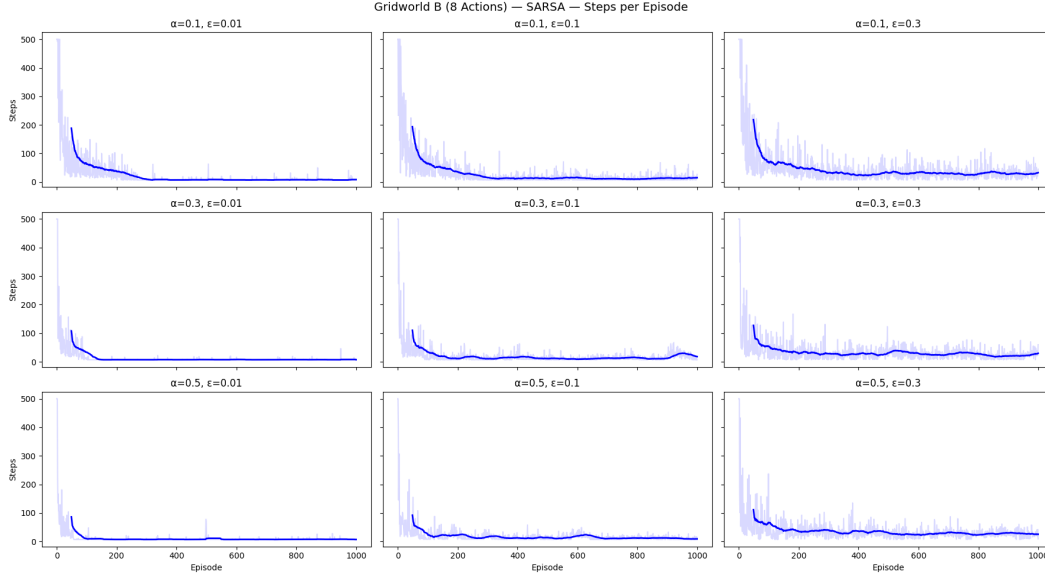


Figure 5: Learning curves (steps per episode) for Gridworld B across all combinations of  $\alpha$  and  $\epsilon$  with 8 actions

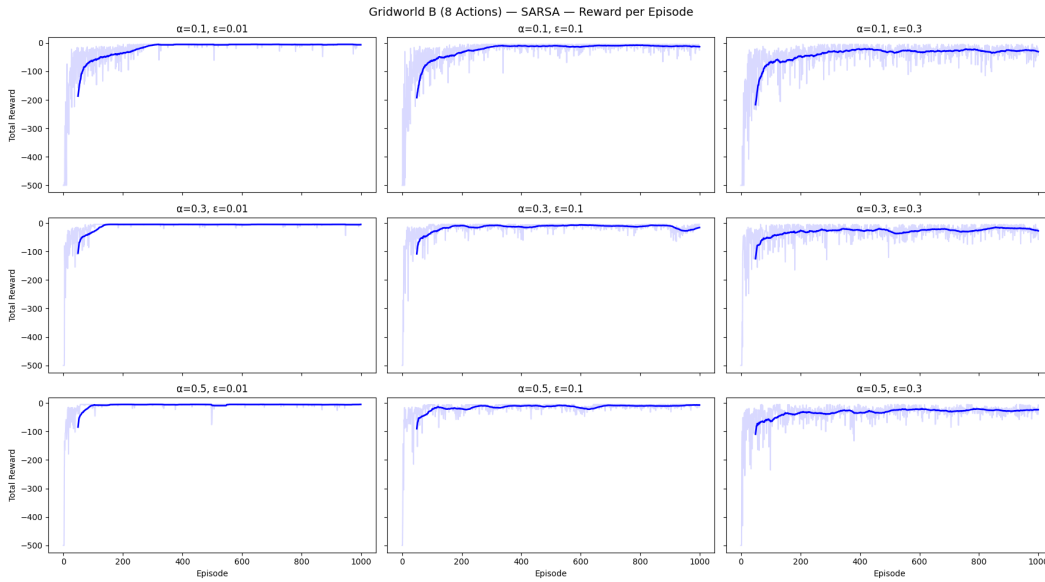


Figure 6: Learning curves (reward per episode) for Gridworld B across all combinations of  $\alpha$  and  $\epsilon$  with 8 actions

123 step. Ultimately, while the 8-action SARSA algorithm can significantly outperform the 4-action  
 124 version when carefully tuned, it is much more sensitive. It specifically needs a lower exploration rate  
 125 to consistently benefit from the added diagonal moves.

Table 7: Convergence Time, Path Efficiency, and Robustness – SARSA with 8 Actions (Gridworld B)

| $\alpha$ | Convergence (episodes) |                     |                     | Path Efficiency (avg steps) |                     |                     |
|----------|------------------------|---------------------|---------------------|-----------------------------|---------------------|---------------------|
|          | $\varepsilon = 0.01$   | $\varepsilon = 0.1$ | $\varepsilon = 0.3$ | $\varepsilon = 0.01$        | $\varepsilon = 0.1$ | $\varepsilon = 0.3$ |
| 0.1      | 202                    | 190                 | 276                 | 7.00                        | 7.00                | 500.00              |
| 0.3      | 94                     | 97                  | 147                 | 7.00                        | 7.00                | 500.00              |
| 0.5      | 63                     | 89                  | 150                 | 7.00                        | 7.00                | 8.00                |

Table 8: Robustness (avg steps and std dev across 10 seeds) – SARSA with 8 Actions (Gridworld B)

| $\alpha$ | $\varepsilon = 0.01$ |      | $\varepsilon = 0.1$ |        | $\varepsilon = 0.3$ |        |
|----------|----------------------|------|---------------------|--------|---------------------|--------|
|          | Avg                  | Std  | Avg                 | Std    | Avg                 | Std    |
| 0.1      | 7.00                 | 0.00 | 7.00                | 0.00   | 106.00              | 197.00 |
| 0.3      | 7.00                 | 0.00 | 106.10              | 196.95 | 106.40              | 196.80 |
| 0.5      | 7.00                 | 0.00 | 10.50               | 3.41   | 109.00              | 195.53 |

Table 9: Direct Comparison: SARSA 4-Action vs 8-Action (Gridworld B)

| $\alpha$ | $\varepsilon$ | Convergence (episodes) |          | Path Efficiency (avg steps) |          |
|----------|---------------|------------------------|----------|-----------------------------|----------|
|          |               | 4-Action               | 8-Action | 4-Action                    | 8-Action |
| 0.1      | 0.01          | 165                    | 202      | 17.00                       | 7.00     |
| 0.1      | 0.1           | 190                    | 190      | 17.00                       | 7.00     |
| 0.1      | 0.3           | 345                    | 276      | 19.00                       | 500.00   |
| 0.3      | 0.01          | 83                     | 94       | 17.00                       | 7.00     |
| 0.3      | 0.1           | 87                     | 97       | 17.00                       | 7.00     |
| 0.3      | 0.3           | 102                    | 147      | 18.00                       | 500.00   |
| 0.5      | 0.01          | 65                     | 63       | 17.00                       | 7.00     |
| 0.5      | 0.1           | 68                     | 89       | 500.00                      | 7.00     |
| 0.5      | 0.3           | 147                    | 150      | 17.00                       | 8.00     |